1.5.3 ▶ ▶ ▶ Kepler's Second Law

Kepler's second law states that the area swept out by a satellite for any duration of time in its orbit is constant. Before I provide a derivation for it, it is important to note that all of Kepler's laws were discovered empirically by Johannes Kepler about ninety years before Newton published his law of universal gravitation. Thus, the following is more of a way to mathematically show that Kepler was correct in his laws, as opposed to a derivation which Kepler himself followed.

To start, consider a small amount of time Δt in which the satellite sweeps out an area ΔA (shown in grey) at a distance r from the star.

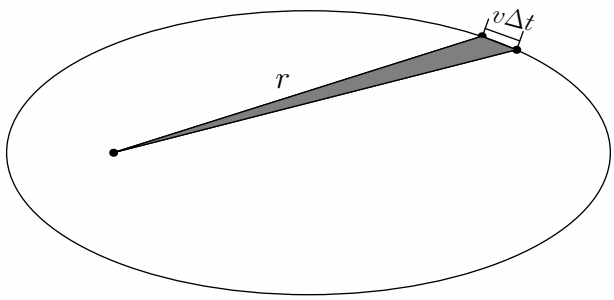

Fig. 1.13

If Δt is small enough, ΔA becomes a triangle[10] with base $v\Delta t$ and height r. Thus, its area is

$$\Delta A = \frac{1}{2} vr\Delta t$$

Rearranging and using the fact that $L = mrv$ gives

$$\frac{\Delta A}{\Delta t} = \frac{L}{2m} \tag{1.77}$$

which is a constant rate since L is a conserved quantity ★

[10] See A.2.1